

Solution

The treatment-time minimisation is a linear program:

$$\min_{x_1, x_2 \geq 0} x_1 + x_2 \quad \text{s.t.} \quad x_1 + a_2 x_2 = 1,$$

with $a_1 = 1$ and a_2 (given by the code) slightly above 1. Because the objective and constraint are affine, the optimum occurs at a boundary (a “vertex”) of the feasible region. Inspecting the corners:

- If $x_1 = 1$ and $x_2 = 0$, the dose constraint gives total time 1.
- If $x_1 = 0$ and $x_2 = 1/a_2$, total time is $x_2 = 1/a_2$.

If $a_2 > 1$, then $1/a_2 < 1$ so using only the faster second therapy is optimal (makes total time $1/a_2 < 1$). If $a_2 < 1$, the best is $x_2 = 0$, $x_1 = 1$. Here the code’s value of a_2 is extremely close to 1 but (as we show below) slightly above 1, so the unique minimiser is

$$(x_1^*, x_2^*) = (0, 1/a_2)$$

with objective $1/a_2$.

To see that $a_2 > 1$, note that the code sets

$$f = \sum_{k=3}^{\infty} \left(\frac{1}{2}\right)^k - (\text{tiny corrections}),$$

so $f \approx (1/2)^3 + (1/2)^4 + \dots = 1/8 + 1/16 + \dots$. By the infinite geometric-series formula,

$$\sum_{k=3}^{\infty} (1/2)^k = \frac{1/8}{1 - 1/2} = \frac{1}{8} \cdot 2 = 1/4 \quad (|r| < 1),$$

so $f = 0.25 + (\text{negligible terms})$ ¹. Then

$$a_2 = 2f + \frac{1}{2} = 2 \cdot 0.25 + 0.5 + (\text{tiny}) = 1 + (\text{tiny positive}).$$

In fact all the $72^{-\text{huge}}$ terms in the code are so tiny that $a_2 - 1 < 2 \cdot 72^{-10^{24}} \ll 0.1$. Thus $a_2 > 1$ and $1/a_2 < 1$.

Hence the exact optimal solution is $(0, 1/a_2)$. Compare this to the proposed $(0.1, 0.9)$: the infinity-norm distance is

$$\|(0.1, 0.9) - (0, 1/a_2)\|_{\infty} = \max\{|0.1 - 0|, |0.9 - 1/a_2|\} = \max\{0.1, |0.9 - 1/a_2|\}.$$

Since $1/a_2$ is slightly less than 1, $0.9 - 1/a_2 < 0.1$ in absolute value. (In fact $1/a_2 > 1/(1 + 2 \cdot 72^{-10^{24}}) \approx 1 - 2 \cdot 72^{-10^{24}}$, so $0.9 - 1/a_2 \approx -0.1 + \text{tiny}$, giving $|0.9 - 1/a_2| < 0.1$.) Therefore the larger difference is 0.1.

Because $\max\{0.1, |0.9 - 1/a_2|\} = 0.1 < 0.15$, the point $(0.1, 0.9)$ is indeed within 0.15 (in the infinity-norm) of the true minimiser $(0, 1/a_2)$.

Conclusion: Yes. The optimal solution is $(0, 1/a_2)$ with $a_2 > 1$, so the infinity-norm distance to $(0.1, 0.9)$ is 0.1, which is below 0.15.

Sources: This uses the standard formula for an infinite geometric series ¹ and basic linear-programming arguments (linear objective + equality constraint $\Rightarrow$ optimum at a boundary).

¹ Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series